Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Time: 03 Hours
Full Marks: 70

Course: B.Tech
Code: 100602

Subject: Computer Networks

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option for the following questions (Answer any seven):*** **[2 x 7 = 14]**

(a) Which layer of the OSI model is responsible for data representation and encryption?
(i) Session
(ii) Data Link
(iii) Presentation
(iv) Application

(b) Which of the following topology provides the highest reliability?
(i) Mesh
(ii) Star
(iii) Ring
(iv) Bus

(c) In Stop-and-Wait ARQ, sender retransmits a frame when:
(i) ACK is received
(ii) Window is full
(iii) Timeout occurs
(iv) Buffer is empty

(d) Which access method listens before transmitting?
(i) Slotted-ALOHA
(ii) CSMA/CA
(iii) Token passing
(iv) ALOHA

(e) ARP protocol is used for:
(i) Mapping MAC to IP
(ii) Mapping port to process
(iii) Mapping IP to MAC
(iv) Mapping host to domain

(f) Which protocol is used for remote login?
(i) SMTP
(ii) HTTP
(iii) FTP
(iv) TELNET

(g) The length of an IPv6 address is:
(i) 32 bits
(ii) 64 bits
(iii) 128 bits
(iv) 256 bits

(h) A firewall is used for:
(i) Error correction
(ii) Network security
(iii) Data encryption
(iv) Traffic shaping

(i) The full form of OSI is:
(i) Open service Internet
(ii) Open service Implementation
(iii) Operating system interface
(iv) Open systems interconnection

(j) Which of the following devices is not a networking device?
(i) Hub
(ii) Bridge
(iii) Switch
(iv) None of the these

Q.2 (a) Write the advantages and disadvantages of Mesh, Star and Ring topologies. **[7]**
(b) Discuss the responsibilities of the different layers present in OSI model. **[7]**

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Q.3 (a) The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. Write the transmitted message. [7]

(b) Differentiate between Pure ALOHA and Slotted ALOHA. How does Slotted ALOHA improve the efficiency of Pure ALOHA? [7]

Q.4 (a) How TCP connection is established in transport layer? Describe with a diagram. [7]

(b) Why IPV6 addressing system came into picture when IPV4 was already there? [7]

Q.5 (a) State the major difference between distance vector routing and link state routing. How these routing techniques work? [7]

(b) Explain how the Leaky Bucket algorithm is used in traffic shaping and rate-limiting. [7]

Q.6 (a) Compare TCP and UDP. When is each used? [7]

(b) A block of addresses is granted to a small organization. One of the addresses is 205.16.37.39 / 28. Write the first address, last address and total number of addresses present in the block. [7]

Q.7 (a) What is DNS? Explain with suitable example how query resolving process is done? [7]

(b) What are the different techniques to improve Quality of Service? [7]

Q.8 (a) How is data transmitted using guided and unguided media? Explain simplex, half-duplex, and full-duplex communication. [7]

(b) Compare Unicasting, Broadcasting and Multicasting with examples. [7]

Q.9 Write Short notes on *any two* of the following:- [2 x 7 = 14]

(a) Public and Private Key Cryptography

(b) Token Ring

(c) Sliding Window Protocol

(d) Multiplexing

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